

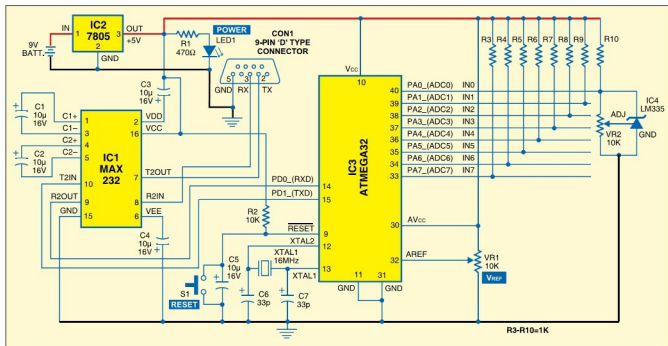


Fig. 3: Circuit for eight-channel data acquisition and logging

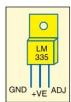


Fig. 4: Pin details of LM335

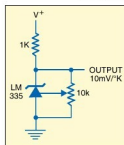


Fig. 5: Circuit for calibration of LM335 to 2.982V at 25°C

any kind of temperature sensing application over the temperature range of -55°C to 150°C . Low impedance and linear output make it easier to interface with the readout and control circuitry. It is not internally calibrated

for degree Celsius ($^{\circ}\text{C}$), so you need some external circuitry in the form of a 10-kilo-ohm preset and a 1-kilo-ohm pull-up resistor as shown in Fig. 5.

Calibration. Calibration is done carefully to map voltage values exactly into temperature in degree Celsius. Calibration procedure is simple. Voltage values are measured for different temperatures and a constant multiplying factor is obtained. This constant is multiplied with the current ADC value every time.

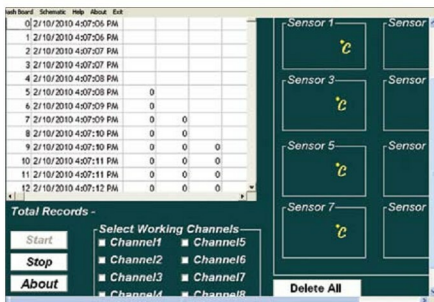


Fig. 6: GUI program output on the dashboard

When calibrated at 25°C , typically, LM335 has an error of less than 1°C over a range of 100°C . Most of all, it has a linear output. The voltage across the output terminal of LM335 is 2.982V at 25°C .

This microcontroller works with TTL digital logic, while the RS-232 standard specifies different voltage

levels of the digital logic. So you need a signal-level converter for communication between the microcontroller and the PC over RS-232 port.

Signal-level conversion. MAX232 is used as the signal-level converter. For voltage-level conversion, four electrolytic capacitors ($10\mu\text{F}$, 16V) are used with MAX232.